

Industrial Equipment Failure Prediction Using Edge Analytics

Real-time predictive maintenance monitoring for Industrial Asset.

Current Asset: Heavy-Duty Liquid Cooling Pump Motor.

Industrial Equipment Failure Prediction Using Edge Analytics

ML approach using Isolation Forest & Linear Regression

Course: Major Project | MCA – AI/ML

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Vibration RMS

2.958 mm/s

Casing Temp

49.96 °C

Motor Current

16.07 A

Coolant Level

10.04 cm

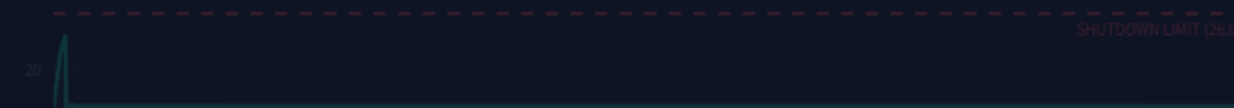
Vibration Trends [Normal]



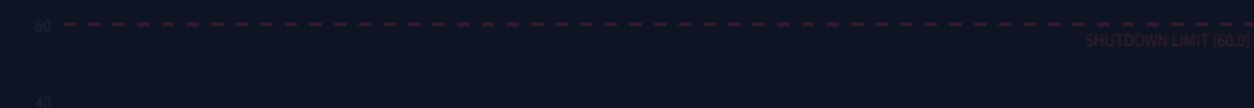
Temperature Profile [Normal]



Motor Current [Normal]



Coolant Tank Ultrasonic Distance [Normal]

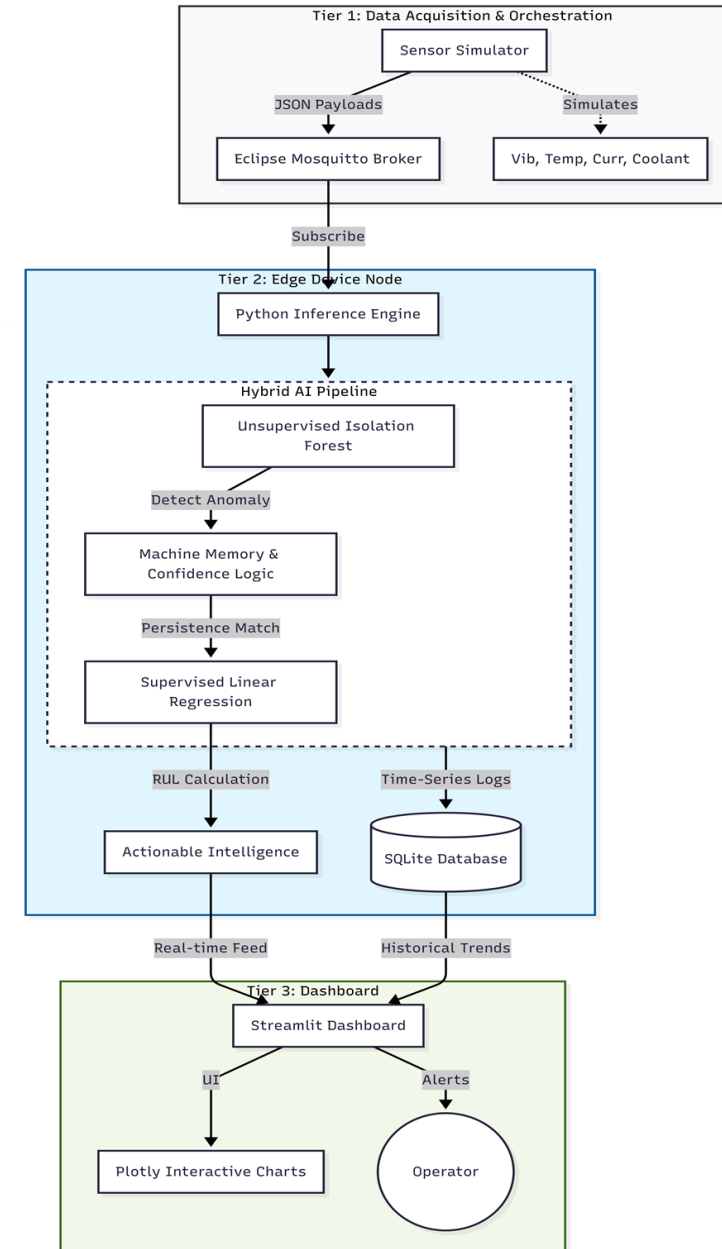


SYSTEM OVERVIEW

Problem Statement

Existing System like Standard SCADA systems rely on static thresholds, often missing the subtle "drift" of mechanical degradation. Whereas the Cloud based compute creates latency risk, bandwidth inefficiency and connectivity blind-spots. Our solution employs a **hybrid** methodology at the edge device using following:

- **Deterministic Layer:** Statistical Z-Score overrides for sudden spikes.
- **Probabilistic Layer:** Unsupervised Isolation Forest for multi-vector anomaly detection.
- **Prognostic Layer:** Linear Regression for real-time RUL forecasting.



USE CASE ANALYSIS



UC-01: ML Training

Initializes "Machine Memory" from SQLite history; fits Isolation Forest model on healthy operational data baseline.



UC-02: Evaluation

Real-time ingestion of telemetry to assess machine state against unsupervised ML boundaries.



UC-03: RUL Predict

Activates supervised Linear Regression upon anomaly detection to project Remaining Useful Life based on degradation velocity.

Primary Actors: Sensor Module (System), Edge AI Node (System), Maintenance Engineer (Human).

HYBRID ANOMALY DETECTION

A. Isolation Forest

Identifies multi-vector outliers by isolating data points through decision trees. Anomalies require fewer splits.

Hyperparameter: *contamination=0.02* (2% historical noise assumption).

B. Z-Score Override

Parallel statistical math to catch rapid linear degradations and prevent "drift blindness."

$$Z = \frac{X - \mu}{\max(\sigma, 0.5)}$$

Override Threshold: $|Z| > 3.5$

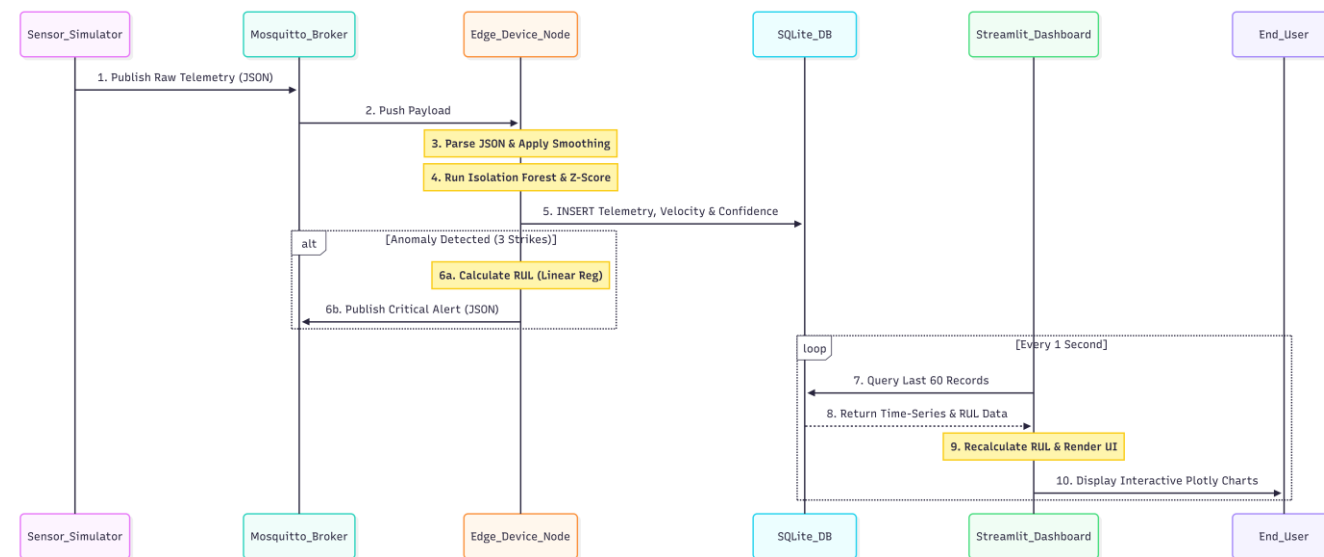
RUL FORECASTING

Linear Regression Engine

Once a fault is latched (3-strike persistence), the system buffers the last 10 seconds of data.

$$RUL = \frac{\text{SafetyLimit} - \text{CurrentValue}}{\text{ProjectedVelocity}}$$

The **slope** of the regression hyperplane determines the degradation velocity, providing a deterministic countdown to the NEMA/ISO safety limit.



FINITE STATE MACHINE (FSM)

State	Name	Logic & Trigger Condition
State 0	Initialization	Boots MQTT, trains baselines, applies Poison Filter to SQLite history.
State 1	Rest/Monitoring	Nominal operation. Consecutive anomaly counter is zero.
State 2	Drift Identification	Unverified anomaly (Strike 1 or 2). System holds for confirmation.
State 3	Stage 1 Warning	Latched alarm (Strike 3+). RUL Engine active. Digital Twin alerts.
State 4	Stage 2 Critical	Hard Limit (e.g. >80°C) breached. Emergency Shutdown (ESD) issued.
State 5	Repair Validation	Hysteresis logic: Requires 3 consecutive healthy readings to return to State 1.

UML SEQUENCE INTERACTION

1. Perception

Sensor Simulator publishes Raw Telemetry (JSON) to Mosquitto Broker at 1 Hz.

2. Intelligence

Edge AI Node parses payload, applies Smoothing, runs Hybrid Inference, and logs to SQLite.

3. Presentation

Streamlit Dashboard polls SQLite asynchronously, renders Plotly charts and RUL countdown.



Latency Optimized: End-to-end processing < 80ms.

JSON MESSAGING SCHEMA

Diagnostic Payload

Upon fault identification, the system dynamically generates Root-Cause Analysis (RCA) metadata.

- Status & Message tags.
- Faulty Sensor ID.
- Prognostic Confidence markers.

```
{ "status": "CRITICAL", "message": "LIMIT  
EXCEEDED: temperature", "prognostics": {  
  "confidence_pct": 94.5, "estimated_rul_sec": 42 },  
  "sensor_data": { "vibration_rms": 3.102,  
    "faulty_sensor": "temperature" } }
```

VALIDATION & PERFORMANCE



Unit Testing

Verified +1.0°C/sec slope detection and Gaussian noise filtering (< 2% variance).



Integration

Validated concurrent SQL read/write and 1 Hz MQTT throughput with zero database lockout.



Performance

Avg Latency: **30-80ms** (Exceeding NFR-01 target of 100ms).

DEPLOYMENT MANUAL

- 1. Environment:** Run `pip install -r requirements.txt` to setup ML and MQTT frameworks.
- 2. Connectivity:** Start Mosquitto Broker on Port 1883.
- 3. Simulation:** Launch `python cooling_pump_sim.py` (1 Hz telemetry).
- 4. Intelligence:** Launch `python edge_node.py` (Isolation Forest / Z-Score).
- 5. Dashboard:** Launch `streamlit run dashboard.py` for real-time visualization.

Master Command: `python run_all.py`

REFERENCES

- [1] ISO 20816-1:2017 Mechanical vibration standards.
 - [2] NEMA MG 1-2021: Motors and Generators guidelines.
 - [3] Liu et al. (2008) "Isolation Forest" IEEE ICDM.
 - [4] Si et al. (2011) "Remaining useful life estimation" EJOR.
- and more...

Questions?

Thank you for your attention.

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github.com/smarthsalaria/predictive_maintenance